SimpsonEF: EXPLAINABLE EJECTION FRACTION FROM PREDICTED SONOGRAPHER TRACINGS AND SIMPSON'S RULE

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ABSTRACT

Video networks estimate left-ventricular ejection fraction (EF) accurately but return a number whose derivation cannot be checked. SimpsonEF computes EF as a sonographer does: it draws the volume tracing (long axis and 20 chords) at end-diastole (ED) and end-systole (ES) of every beat and applies Simpson's method of disks, with no learned EF regressor. FrameTracer traces every frame, BeatFinder finds the beats on the resulting volume curve, and BeatTracer draws the ED and ES tracings of each beat jointly, trained through Simpson's formula in EF units. On all 1276 EchoNet-Dynamic test videos, with no annotated frame at inference, formula EF reaches R^2 0.808 and MAE 4.00 (3.4M parameters, trained from scratch), against 0.65 for keypoint-derived EF under the same protocol. We trace the compression of low and high EF towards normal to the training objective and to input-noise augmentation, and give conformal intervals (90.6% coverage). Zero-shot on CAMUS, biplane formula EF reaches R^2 0.710.

Index Terms— Ejection fraction, echocardiography, Simpson's method, keypoints, explainability, conformal prediction

1. INTRODUCTION

Left-ventricular ejection fraction (LVEF, here EF), the fraction of the end-diastolic volume ejected in a beat, is the principal measure of systolic function [1]. Video networks estimate EF to about 4 EF points [2, 3] but return a bare number. Clinicians compute EF from a tracing by Simpson's method of disks, so a model whose output is that tracing, and whose EF is that formula, can be checked frame by frame.

Related work. Direct regression with 3D CNNs [2], CNN–transformers [4], contrastive regression [3] or light spatio-temporal networks [5] reaches R^2 0.81–0.82. Foundation models pretrained on millions of echocardiograms [6, 7, 8, 9] give strong EF probes but are 28M–1.1B-parameter black boxes. Explainable models use graph attention [10] or prototypes [11] but still regress EF; uncertainty-aware ones use Bayesian networks [12] or conformal prediction [13]. Keypoint models come closest to the clinical computation: EchoTrace [14] regresses the 84 tracing coordinates of single frames, and EchoGraphs [15] finds contour keypoints with a graph network but reports that Simpson's EF from them (R^2 0.65 on whole videos) trails a black-box regressor (0.79).

We contribute: (i) a formula-based pipeline on whole videos with every intermediate visible; (ii) joint ED/ES tracing trained through the formula in EF units, closing most of the gap to direct regression; (iii) an analysis of where formula EF loses accuracy; and (iv) conformal intervals and a zero-shot external test.

2. METHOD

Tracings and EF. A tracing has the ends A_1, A_2 of the long axis and the ends L_i, R_i of 20 perpendicular chords (EchoNet-Dynamic convention). With $d_i = \|L_i - R_i\|$ and $\ell = \|A_1 - A_2\|$,

$$V = \frac{\pi}{4}\,\frac{\ell}{20}\sum_{i=1}^{20} d_i^2, \qquad \mathrm{EF} = 100\Big(1 - \frac{V_{\mathrm{ES}}}{V_{\mathrm{ED}}}\Big). \tag{1}$$

No mask is a training target or enters the EF; masks only measure Dice.

Tracers. FrameTracer and BeatTracer are fully convolutional encoder–decoders (instance norm, GELU, skip connections, dilated bottleneck, no global pooling) with one full-resolution heatmap per point; each point is the soft-argmax [16] over a 7×7 window at the peak. The same weights run at scales 1, 1/2 and 1/4 and the logits are fused by log-sum-exp, so Dice changes by less than 0.01 between 0.5× and 3× the training size (single-scale: 0.54 at 3×). FrameTracer (1.42M parameters) takes 13 frames (offsets −24..24, step 4) and returns 42 points. BeatTracer (1.43M) takes the 25-frame windows (step 2) around ED and around ES stacked as channels and returns both tracings (84 points), so ES is drawn knowing the ED appearance.

Beats. BeatFinder, a two-layer bidirectional GRU (0.53M), reads FrameTracer's chord sequence $x_t = (d_1, \ldots, d_{20}, \ell)$ and outputs ED and ES heatmaps over time (Gaussian targets around the annotated beat); peaks give each beat's ED and following ES. It only chooses frames, which can be read off the volume curve.

Training through the formula. Both tracers use an L_1 point loss, heatmap cross-entropy (σ=1.5 px) and L_1 losses on $\log V$, on its ED→ES change and on volume-weighted log chord widths, with similarity transforms (rotation ±20°, scale 0.8–1.2, shift ±18 px, canvas 0.5–2.5×), edge occlusion and scanner-style degradations (gain, gamma, speckle, noise, blur, shadows). The change of $\log V$ is $\log(1-\mathrm{EF})$, so an EF point costs $\approx 1/(1-\mathrm{EF})$: 1.4 at EF 30% but 3.3 at 70%, i.e. this loss values low-EF hearts least. BeatTracer is therefore fine-tuned with $\mathscr{L}_{\mathrm{EF}} = w_b\,|\mathrm{EF}(\hat{P}_{\mathrm{ED}}, \hat{P}_{\mathrm{ES}}) - \mathrm{EF}^*|$ in EF units, w_b the inverse square root of the frequency of the label's 10-point EF bin (15 epochs), then without dropout and canvas rescaling (10), then without scanner noise (20): input noise shrinks outputs towards the mean (attenuation bias [17]); removing it only at the end keeps the robustness it brings.

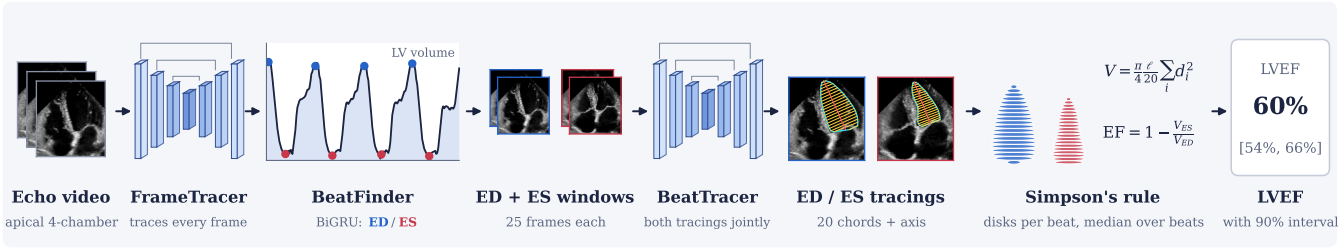


Fig. 1. SimpsonEF on a TEST case. FrameTracer (an encoder–decoder) traces every frame; BeatFinder marks ED and ES of every beat on the resulting volume curve; BeatTracer reads the frame windows around a beat's ED and ES together and draws both tracings; Simpson's rule turns the 20 chords into disks and volumes. LVEF is the median over beats, with a conformal 90% interval.

Inference and uncertainty. BeatTracer is applied to 15 rescaled (0.85–1.17) and rotated (±8°) copies of each beat's input and the mapped-back points are averaged; (1) gives the beat's EF and the median over beats is reported. A normalised split-conformal interval [18] uses the scale $s = (\hat{q}_{.95} - \hat{q}_{.05})/2 + |\widehat{\mathrm{EF}} - \widehat{\mathrm{EF}}_{\mathrm{GRU}}|$, where BeatFinder also predicts EF quantiles from FrameTracer's chords: s grows when the clip is uncertain or the two routes to EF disagree. With Q the conformal quantile of $|y - \widehat{\mathrm{EF}}|/s$ on VAL, $\widehat{\mathrm{EF}} \pm Qs$ covers 90% of exchangeable cases.

Interpretability. No learned component maps an image to EF. Every estimate decomposes into quantities a clinician can inspect and correct: the volume curve and the frames BeatFinder chose, the tracings on those frames, the 20 chord widths and the axis, the arithmetic of (1), the EF of every beat and their median, and the reason for a wide interval (an uncertain clip or two disagreeing routes to EF). A corrected tracing changes EF exactly as in the clinical workflow.

3. EXPERIMENTS

Protocol. EchoNet-Dynamic [2] has 7460/1288/1276 TRAIN/VAL/TEST videos with one annotated beat each; 81 TRAIN records with stacked tracings are excluded from training, all VAL and TEST videos are kept. VAL selects checkpoints and calibrates the intervals; at test time no annotation is used. We report R^2, MAE (EF points) and Pearson r against the EF label, and Dice between the mask drawn from a tracing and the released mask. The tracers are trained from scratch (AdamW, learning rate 3×10^{-4}, cosine decay, EMA 0.999, 32 frames per batch, 40 epochs, then the three fine-tuning stages) and BeatFinder for 16 epochs; one script rebuilds all three networks with fixed seeds, and a full rebuild reproduces the results (R^2 0.807 on TEST, 0.713 on CAMUS). Inference takes 0.49 s per 188-frame video on one RTX 3090 (0.23 s without input-copy averaging).

Accuracy. Formula EF reaches R^2 0.808, MAE 4.00 and r 0.901 on all TEST videos (Table 1): 0.16 above keypoint-derived EF under the same protocol, within 0.01–0.02 of video regressors, and with lower MAE than all foundation models except Echo-Vision-FM and the 1.1B-parameter EchoJEPA-G. BeatTracer's tracings reach Dice 0.916/0.907 (ED/ES). Table 2 shows the steps; the EF-unit formula loss is the largest gain, and Simpson's formula on BeatTracer's tracings outperforms a learned EF head on the same pipeline's chords (0.799). Regressing the 84 coordinates from a pooled feature, as in EchoTrace, instead of heatmaps lowers Dice from 0.915/0.905 to 0.885/0.871 (VAL).

Table 1. EF on EchoNet-Dynamic TEST as reported by each work (protocols differ; whole videos unless marked). All but ours and two EchoGraphs rows regress EF; formula: Simpson's rule on predicted geometry. *Annotated beat given. †Frozen backbone and probe, from [9]. §External Stanford cohort. ¶Backbone only (total not reported). n.r.: not reported.

Method	Par. (M)	MAE	R^2	EF from
Video regression				
EchoNet-Dynamic [2]	32	4.05	0.81	regression
EchoCoTr-B [4]	21	3.98	0.81	regression
CoReEcho [3]	21	3.90	0.82	regression
Echo-E^{3}Net [5]	1.54	3.93	0.82	regression
Foundation models				
Echo-Vision-FM [8]	86¶	3.87	0.825	regression
EchoPrime§ [6]	35	4.14	0.79	regression
EchoJEPA-G† [9]	1100	3.97	n.r.	regression
PanEcho† [7]	28	5.10	n.r.	regression
Explainable / uncertainty-aware				
Bayesian [12]	31.5	4.46	0.75	regression
EchoGNN [10]	1.7	4.45	0.76	regression
ProtoEFNet [11]	31.5¶	4.07	0.80	regression
Keypoints / tracings				
EchoGraphs [15]	27.6	4.23	0.79	regression
EchoGraphs, given beat*	27.6	4.66	0.73	formula
EchoGraphs	27.6	4.30	0.65	formula
SimpsonEF (ours)	3.38	4.00	0.808	formula

Where the error is. Simpson's rule on the sonographer's own TEST tracings reproduces the label (R^2 0.943), so the error is in the predicted tracing, and there in the chord widths: true widths give 0.930, true axis lengths barely help. Other functions of the chords (area–length volumes, learned weights, boosted trees) stay at $r \approx 0.88$. The residual is EF compressed towards normal (slope 0.81; Fig. 3a): +3.6 to +5.5 points below EF 50%, ±1.2 in 50–70%, −5.7 above 70%. It appears on TRAIN too (slope 0.83), i.e. underfitting, which the EF-unit loss and removing input noise address; reweighting the tails flattens it (0.86) but costs 0.01 in R^2.

Uncertainty. Calibrated on VAL, the intervals cover 90.6% of TEST cases (95% CI 88.9–92.1) with mean width

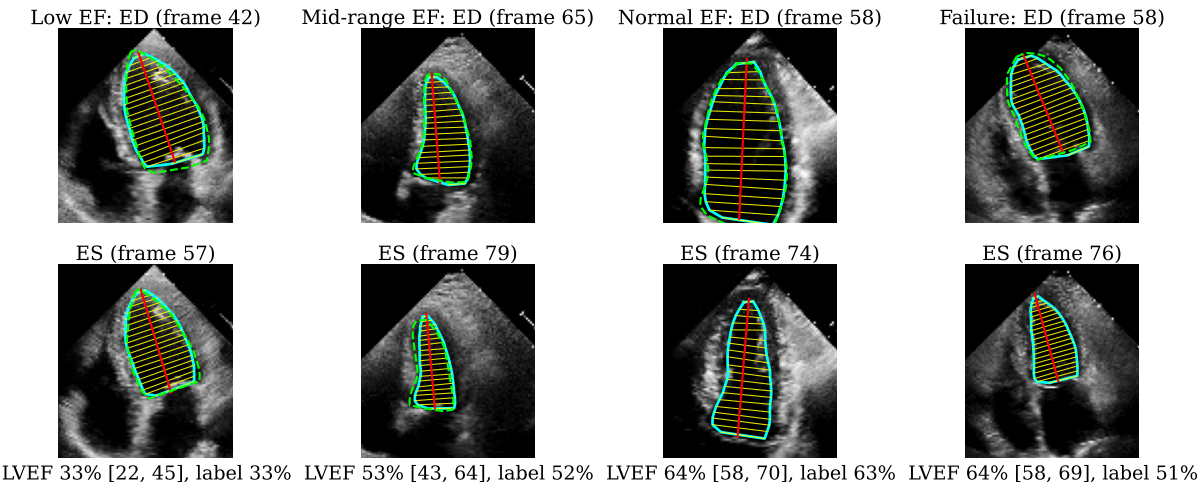


Fig. 2. TEST cases: ED (top) and ES (bottom) with BeatTracer's tracing (cyan, chords yellow, axis red) and the reference (green dashed); formula EF with its 90% interval and the label. Left to right: low, mid-range and normal EF, and a failure (label outside the interval).

Table 2. Ablations: formula EF on all 1276 TEST videos, beats from BeatFinder. *With a single fine-tuning stage (reference 0.804).

Variant	R^2	r	slope
FrameTracer tracings only	0.768	0.879	0.80
+ joint ED/ES tracing (BeatTracer)	0.773	0.882	0.76
+ EF-unit formula loss	0.790	0.892	0.80
+ input-copy averaging	0.798	0.896	0.81
+ no dropout / no scanner noise	0.804	0.899	0.82
+ 25-frame windows (SimpsonEF)	0.808	0.901	0.81
One change to SimpsonEF			
learned EF head instead of Simpson	0.799	0.894	–
no input-copy averaging	0.801	0.896	0.81
mean instead of median over beats	0.780	0.883	–
tail-oversampled EF stage*	0.792	0.894	0.86
wider tracer (width 32)*	0.794	0.894	0.81

16.5, and 88% of cases below EF 40% (constant width: 71%). Width tracks error: it detects errors above 10 points with AUROC 0.75, and excluding the 20% widest intervals lowers the MAE of the rest from 4.00 to 3.52 (Fig. 3b). Beat-to-beat spread alone was a weaker scale (AUROC 0.62).

Zero-shot CAMUS. CAMUS [19] differs in protocol, not only images: (i) its EF label is biplane Simpson (4- and 2-chamber views) with one disk height shared by ED and ES, whereas EchoNet labels are single-plane with per-phase heights; (ii) each view is a half cycle ED→ES (≈20 frames), not a multi-beat video; (iii) contours follow another convention; (iv) images are native-resolution sectors (we pad to a square and resize to 112). We run FrameTracer and BeatFinder unchanged on the 50 test patients, with no CAMUS data in training and no annotated frame, and apply the CAMUS label definition to the chords (its reference code

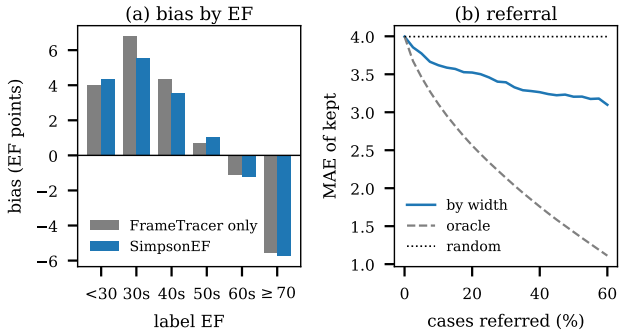


Fig. 3. TEST: (a) bias by label EF, FrameTracer alone and SimpsonEF; (b) MAE of the remaining cases when those with the widest intervals are excluded.

reproduces the label from the reference masks, R^2 0.998). BeatTracer, trained on full-beat windows, transfers worse (R^2 0.61).

Biplane formula EF reaches R^2 0.710 (Table 3; Dice 0.905/0.868 4CH, 0.893/0.864 2CH); with EchoNet's per-phase disk heights the same chords give $R^2 \approx 0$ (bias +9), so the label convention matters as much as the model. The rest is contour convention (Fig. 4): our contours are narrower towards the base, most at ES; the reference 4-chamber chords alone would give 0.79. A zero-shot model cannot learn this convention.

4. DISCUSSION AND CONCLUSION

SimpsonEF obtains EF from tracings and Simpson's rule on whole videos with the accuracy of video regressors and 3.4M parameters, so every estimate can be checked on the frames that define it, and its interval quantifies how far it can be trusted. Limitations: low and high EF stay biased by about

Table 3. Zero-shot CAMUS test (50 patients): formula EF from FrameTracer's chords at the detected ED/ES, CAMUS disk-height convention.

EF definition	R^2	MAE	r
Biplane, one disk height (label)	0.710	5.08	0.849
4-chamber only	0.567	5.77	0.804
2-chamber only	0.592	6.20	0.788

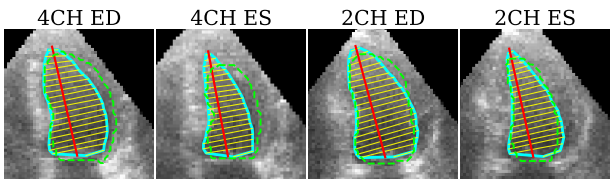


Fig. 4. Zero-shot CAMUS: FrameTracer's tracing at the detected ED/ES of each view and the CAMUS reference (green dashed), wider towards the base.

5 points; beat finding uses a second tracer (folding it into BeatTracer gave 0.789); EF is single-plane, while guidelines prefer biplane; CAMUS conventions cannot be learned zero-shot; ED/ES supervision is one beat per video; no reader study. Future work: biplane EF with the same tracers, multi-dataset training with an explicit convention input, single-network beat finding, attenuation-bias correction that keeps noise robustness, and prospective evaluation. Code: https://github.com/gowthamaanp/shape-ef.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used open-access human subject data from EchoNet-Dynamic (Stanford University) and CAMUS (University Hospital of St Etienne); ethical approval was not required as confirmed by the licences attached to the data.

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